PARTICIPANT DATA

Sample Size
26

Location
New York, USA

Age Distribution

Over 26 Under 26

Mean
26.31
(Range: 18 – 57)

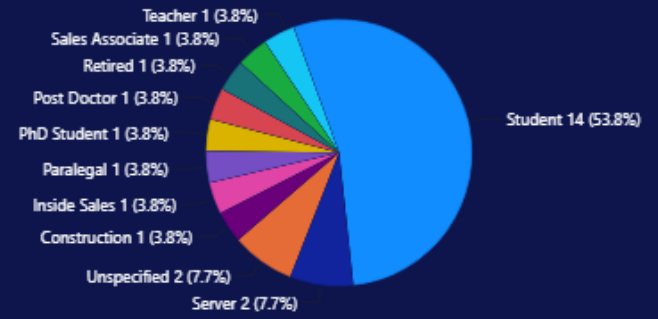


Sex Distribution

Female Male



Occupation Distribution

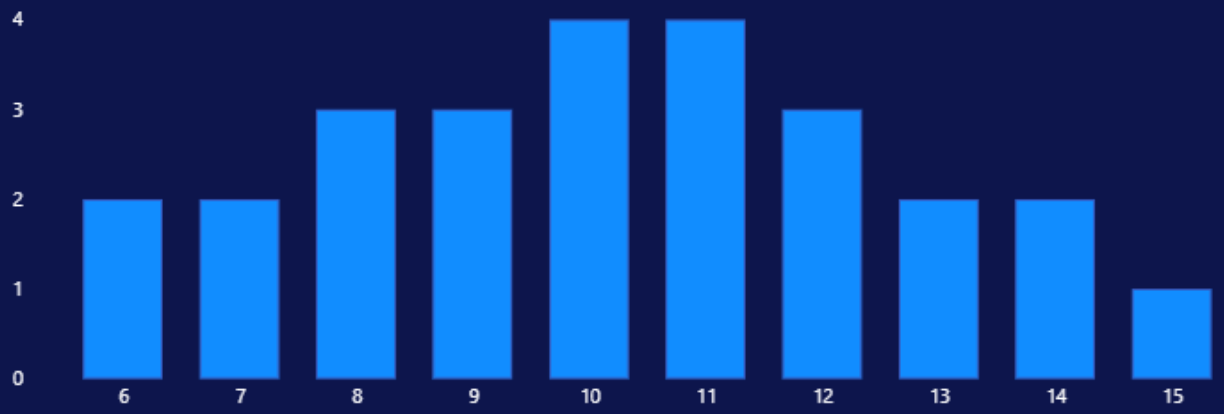


ASRS Test Results

Score Distribution

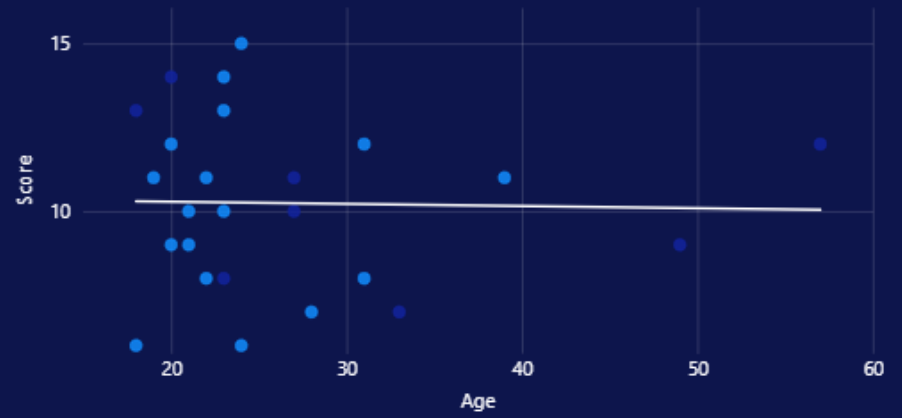
Mean
10.23

ASRS | Digit Span



ASRS Score by Age and Sex

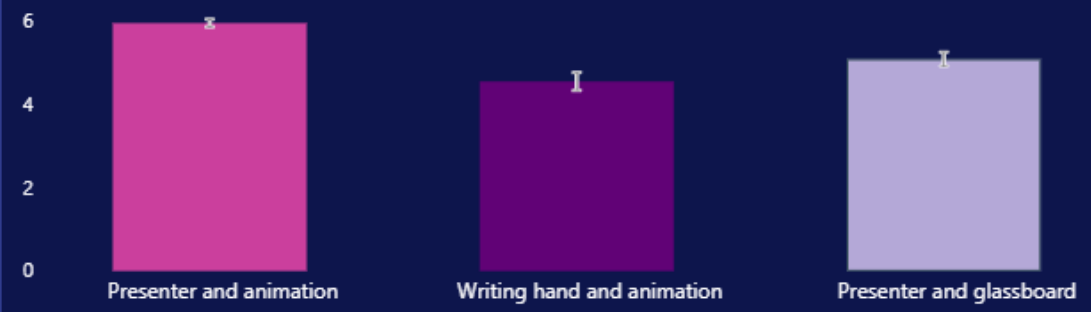
Female Male



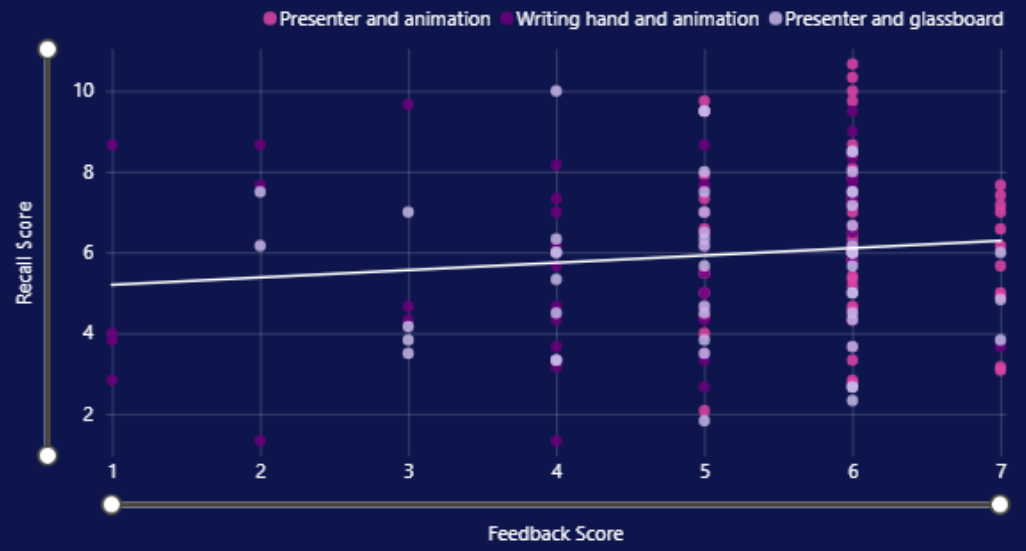
BEHAVIOURAL RESULTS

Stimuli Feedback
Mean Ratings by Attribute

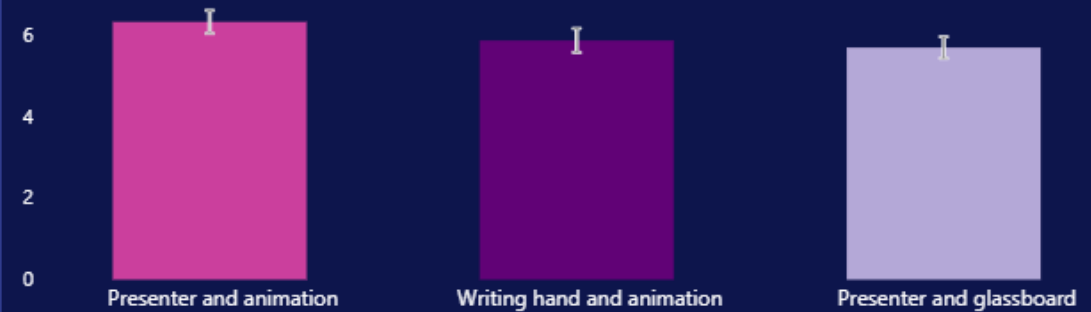
Educational | Engaging | Enjoyable



Stimuli Feedback x Recall
Score Association

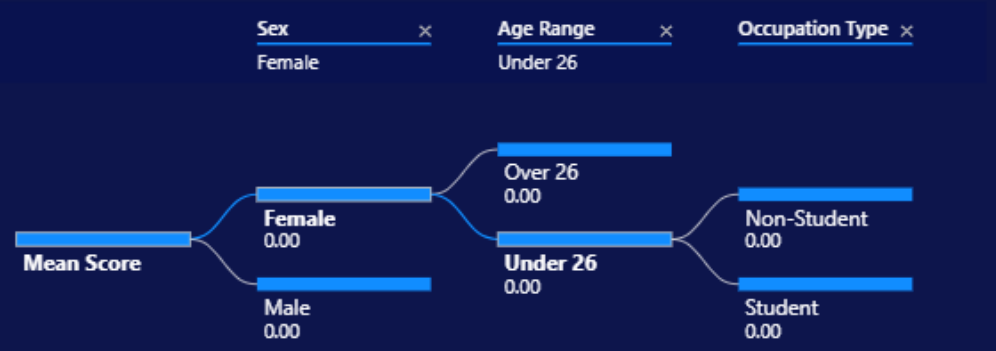


Stimuli Recall Quiz
Mean Accuracy Scores



Stimuli Feedback - Decomposition Tree
Mean Centred Score by Demographic Segment

Feedback | Recall



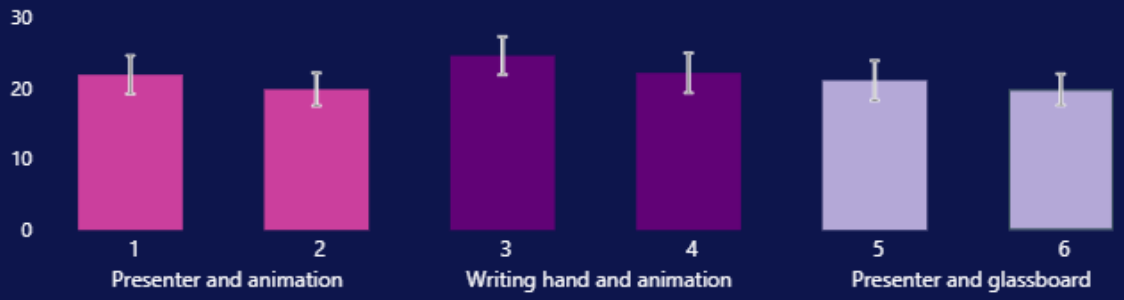
EYE TRACKING RESULTS

Attentive | Distracted

Stimuli Comparison

Mean Metrics

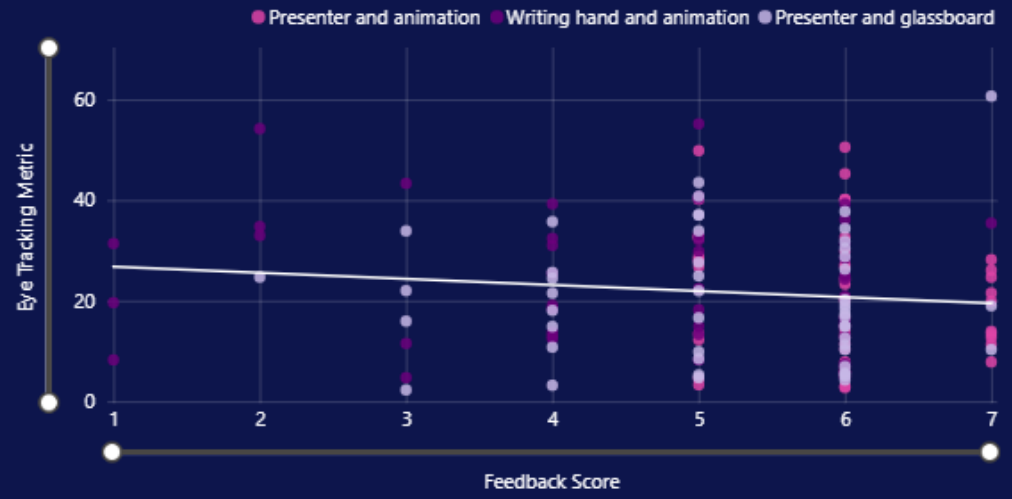
Blink Rate | Fixation Duration | Fixation Rate



Stimuli Feedback x Blink Rate

Score Association

Educational | Engaging | Enjoyable

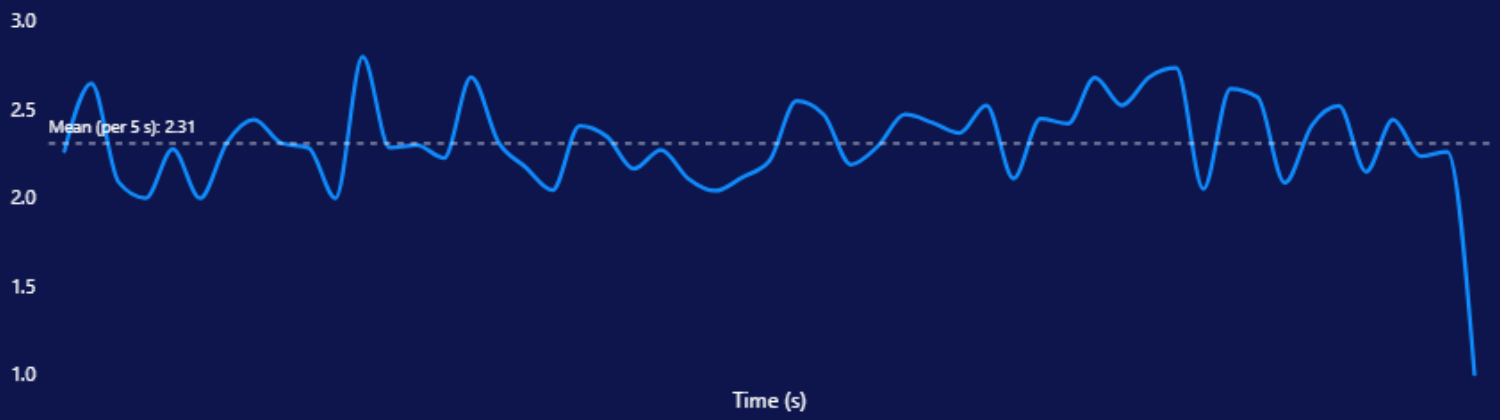


Temporal Dynamics

Metrics over Time

Blink Count | Fixation Count | Fixation Duration

Stimulus: 1



5 Peak Moments

Metric Value	Timestamp
2.80	00:55
2.74	03:25
2.68	01:15
2.68	03:20
2.68	03:10

5 Lowest Moments

Metric Value	Timestamp
1.00	04:20
2.00	00:15
2.00	00:25
2.00	00:50
2.04	02:00

Link to the Video

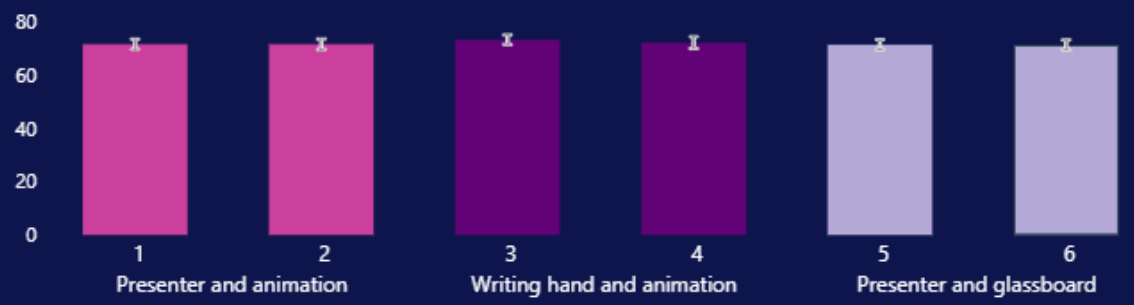
https://www.youtube.com/watch?v=e0NT9i4Qnak

ECG RESULTS

Attentive | Distracted

Stimuli Comparison
Mean Metrics

Heart Rate | RMSSD



Stimuli Feedback x Heart Rate
Score Association

Educational | Engaging | Enjoyable

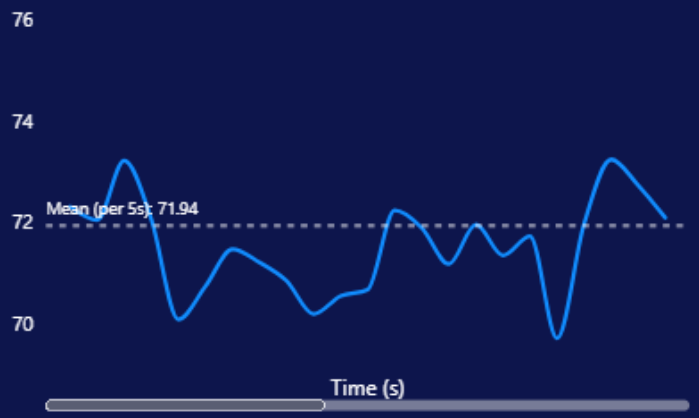


RMSSD - Decomposition Tree
Mean Centred Score by Demographic Segment



Temporal Dynamics
Heart Rate over Time

Stimulus: 1



5 Peak Moments

Heart Rate	Timestamp
75.02	03:10
73.89	03:25
73.87	02:25
73.68	02:15
73.41	02:05

5 Lowest Moments

Heart Rate	Timestamp
69.72	01:30
69.78	03:50
70.05	02:45
70.09	00:20
70.19	00:45

Link to the Video

https://www.youtube.com/watch?v=e0NT9i4Qnak

EEG RESULTS

Attentive | Distracted

PSD Condition Contrast

Power Differences across Sites

Location	Left (3/1)	Central (z)	Right (4/2)
F	0.09	0.31	0.18
C	0.77	0.29	0.76
P	-0.01	-0.02	0.36
O	0.26		0.38

Contrast: Presenter and animation - Presenter and glassboard

Frequency Band: Alpha

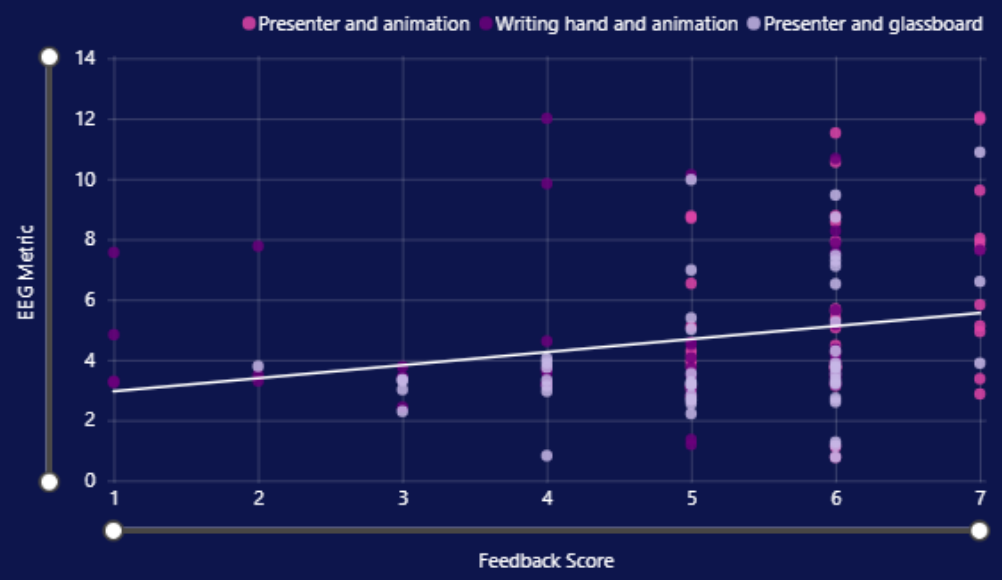
Mean Contrast per Cluster

F	0.19
C	0.07
P	0.33
O	0.27

Stimuli Feedback x Central Theta Power

Score Association

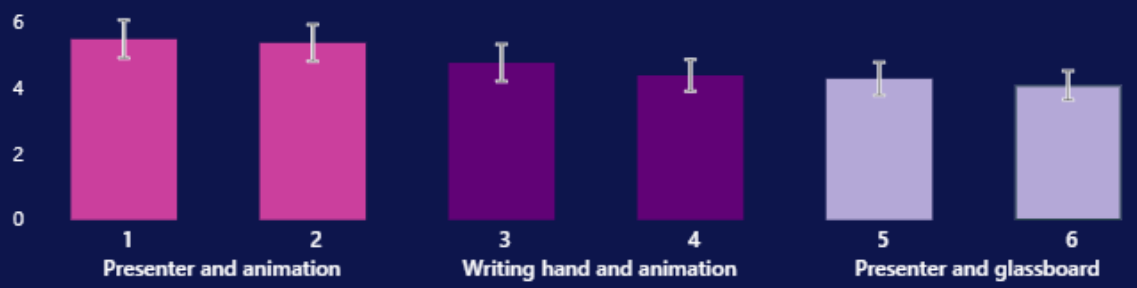
Educational | Engaging | Enjoyable



Stimuli Comparison

Mean Metrics

Central Theta Power | Frontal Alpha Asymmetry | Frontal Alpha Power



EEG Metrics x Feedback (Educational)

Spearman Correlations

EEG/Score	Frontal Alpha Asymmetry		Frontal Alpha Power		Frontal Theta Power	
Metric	Average of Spearman's Rho	Average of P value	Average of Spearman's Rho	Average of P value	Average of Spearman's Rho	Average of P value
Feedback Score	-0.03	0.765	0.14	0.110	0.31	0.001
Frontal Theta Power	0.19	0.028	0.71	0.001		
Frontal Alpha Power	0.18	0.037				

Sex

- [ ] Select all
- [ ] Female
- [ ] Male

Age Range

- [ ] Select all
- [ ] Over 26
- [ ] Under 26

Occupation Type

- [ ] Select all
- [ ] Non-Student
- [ ] Student

ASRS Score

- [ ] Select all
- [ ] Above Average
- [ ] Below Average

Digit Span Score

- [ ] Select all
- [ ] Above Average
- [ ] Below Average

Attentive | Distracted

S

(z)	Right (4/2)
	0.18
	0.76
	0.36
	0.38

assboard

Mean Contrast per Cluster

F: 0.19

C: 0.07

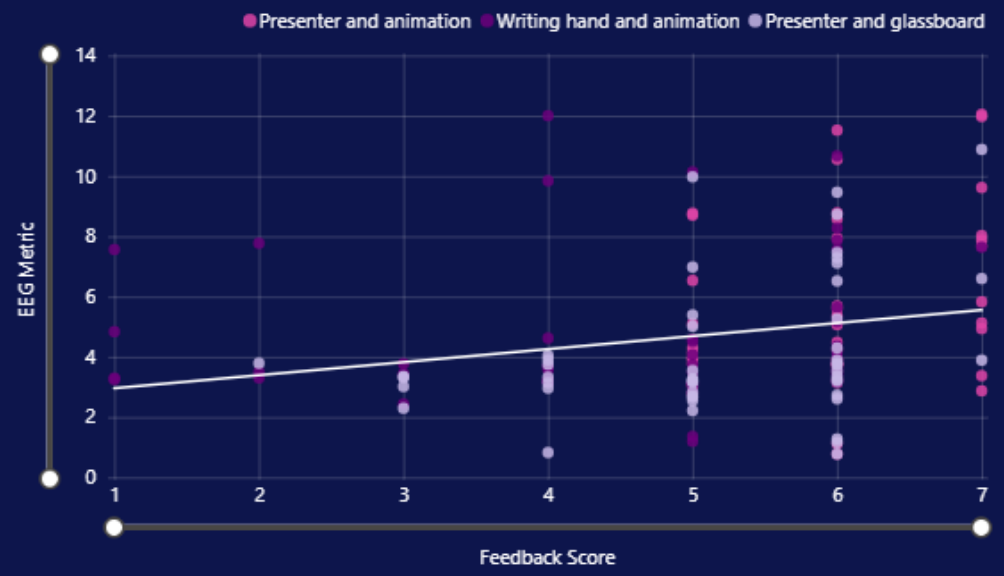
P: 0.33

O: 0.27

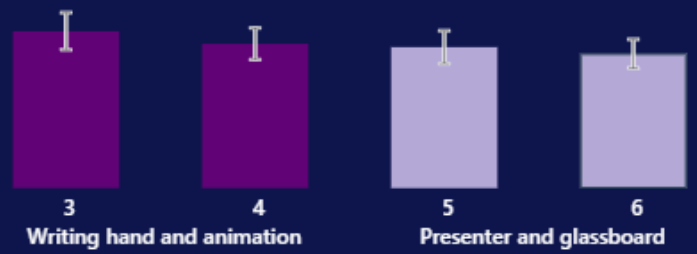
Stimuli Feedback x Central Theta Power

Score Association

Educational | Engaging | Enjoyable



l Theta Power | Frontal Alpha Asymmetry | Frontal Alpha Power



EEG Metrics x Feedback (Educational)

Spearman Correlations

EEG/Score Metric	Frontal Alpha Asymmetry		Frontal Alpha Power		Frontal Theta Power	
	Average of Spearman's Rho	Average of P value	Average of Spearman's Rho	Average of P value	Average of Spearman's Rho	Average of P value
Feedback Score	-0.03	0.765	0.14	0.110	0.31	0.001
Frontal Theta Power	0.19	0.028	0.71	0.001		
Frontal Alpha Power	0.18	0.037				

EEG RESULTS

Attentive | Distracted

PSD Condition Contrast

Power Differences across Sites

Location	Left (3/1)	Central (z)	Right (4/2)
F	0.09	0.31	0.18
C	0.77	0.29	0.76
P	-0.01	-0.02	0.36

Mean Contrast per Cluster

F

0.19

C

0.07

.33

.27

Power

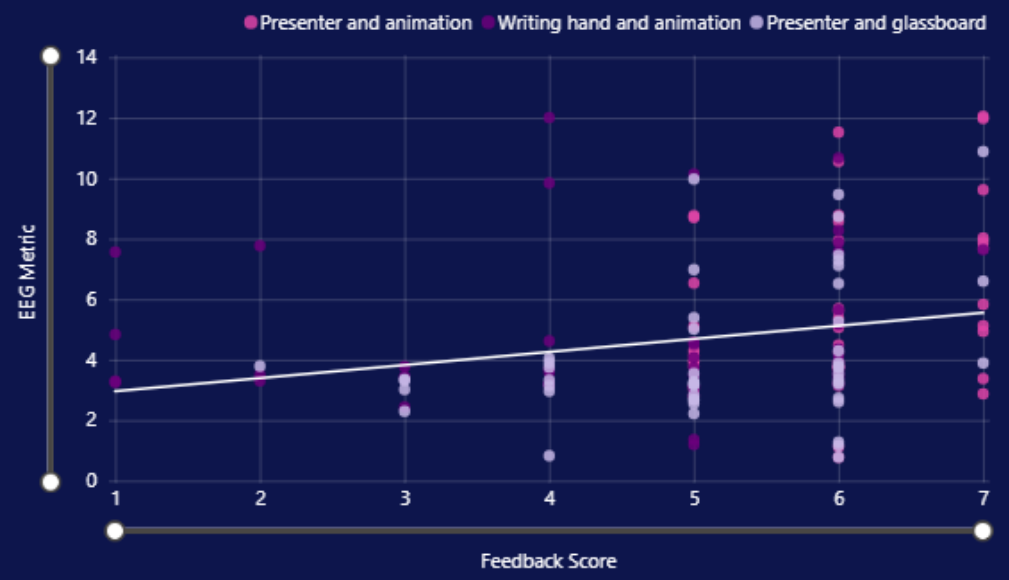
6

and glassboard

Stimuli Feedback x Central Theta Power

Score Association

Educational | Engaging | Enjoyable



Electroencephalography (EEG) measures electrical activity of the brain, offering a window into the neural processes underlying engagement, emotional response, and cognitive effort. Three metrics were analysed:

Frontal Alpha Power - an indicator of the strength of activity within the 8-13 Hz frequency range at a frontal site (Fz), representing relaxation.

Central Theta Power - an indicator of the strength of activity within the 4-8 Hz frequency range at a central site (Cz), utilised as a measure of cognitive effort and working memory engagement.

Frontal Alpha Asymmetry - an index of the balance of activity between the left and right hemisphere, linked to approach- versus withdrawal-related motivation.

EEG Metrics x Feedback (Educational)

Spearman Correlations

EEG/Score	Frontal Alpha Asymmetry		Frontal Alpha Power		Frontal Theta Power	
Metric	Average of Spearman's Rho	Average of P value	Average of Spearman's Rho	Average of P value	Average of Spearman's Rho	Average of P value
Feedback Score	-0.03	0.765	0.14	0.110	0.31	0.001
Frontal Theta Power	0.19	0.028	0.71	0.001		
Frontal Alpha Power	0.18	0.037				